

Conjecture 00000001457 is false

A planar Banach–Mazur obstruction

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Abstract

The conjecture predicts distance $n - 1$ between an n -simplex and the ℓ_1^n unit ball. In dimension two this predicts one. We show that every affine copy of the planar cross-polytope containing the standard triangle requires dilation at least two to fit inside a translated copy of that triangle. The distance is therefore at least two.

1 Bodies and distance convention

Let

$$K = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 1\}, \quad B = \{(x, y) : |x| + |y| \leq 1\}.$$

K is a two-dimensional simplex and B is the cross-polytope, that is, the unit ball of ℓ_1^2 . Use the affine convex-body Banach–Mazur distance

$$d_{\text{BM}}(K, B) = \inf\{\lambda > 0 : K \subseteq c + T(B) \subseteq d + \lambda K, \ c, d \in \mathbb{R}^2, \ T \in \text{GL}_2(\mathbb{R})\}.$$

The middle and outer translations are independent. This convention is equivalent to putting the inner and outer homothetic copies in the usual common-center form when $\lambda > 1$: their homothety center is $d/(1 - \lambda)$. The proof below even allows all positive dilation factors. The multiplicative convention is the one in which the question states its numerical bounds; no logarithm of the distance is being used.

2 A lower bound for every affine image

Theorem 1. *If $K \subseteq c + T(B) \subseteq d + \lambda K$ with $\lambda > 0$ and T invertible, then $\lambda \geq 2$.*

Proof. The middle body is centrally symmetric about $c = (c_1, c_2)$. Since it contains the three vertices $(0, 0), (1, 0), (0, 1)$, it also contains their reflections

$$(2c_1, 2c_2), \quad (2c_1 - 1, 2c_2), \quad (2c_1, 2c_2 - 1).$$

Membership of (x, y) in $d + \lambda K$ is exactly $x \geq d_1, y \geq d_2$, and $x + y \leq \lambda + d_1 + d_2$. Applying the sum inequality to the first reflected vertex, the first coordinate inequality to the second, and the second coordinate inequality to the third gives

$$2c_1 + 2c_2 \leq \lambda + d_1 + d_2, \quad d_1 \leq 2c_1 - 1, \quad d_2 \leq 2c_2 - 1.$$

Combining these three inequalities gives $2 \leq \lambda$. □

The admissible set is nonempty, so taking its infimum is legitimate. For example, choose $T = I$, $c = (1/3, 1/3)$, $d = (-2/3, -2/3)$, and $\lambda = 4$. Each vertex of K lies in $c + B$; convexity gives $K \subseteq c + B$. For any $q \in B$, the coordinates of $c + q$ are at least $-2/3$ and its coordinate sum is at most $5/3 < 8/3$, giving the outer inclusion. Thus the distance lies in $[2, 4]$, which already contradicts the claimed value $2 - 1 = 1$. An exact optimal upper bound is unnecessary. This refutes the statement in a nondegenerate two-dimensional case, without relying on the one-dimensional endpoint. It does not decide a modified conjecture restricted to higher dimensions or an asymptotic claim.

3 Formal verification and semantic correspondence

The Lean plane is $\mathbb{R} \times \mathbb{R}$. `diamond` and `triangle` encode B and K by their defining inequalities. `affineDiamond` uses a real linear equivalence, so it ranges over every invertible real linear transformation and every center. `outerTriangle` is exactly $d + \lambda K$ for $\lambda > 0$: subtract d and divide both coordinates by λ to obtain a point of K ; `outerTriangle_iff` formally verifies this equivalence. `Admissible` encodes precisely the two inclusions above, with independent translations.

`admissible_lower_bound` proves the universal geometric bound; `four_admissible` proves nonemptiness with all inclusions checked pointwise. The real infimum in `distance` is bounded below by two in `distance_lower_bound`, and `conjecture_false` refutes the claimed value one. This is an infimum proof, not merely a failure to attain dilation one.

The project pins Lean and Mathlib. Run `lake build` and `lake env lean Check.lean in lean`. The axiom audit is in `verification.txt`; no admitted proofs, additional axioms, or `native_decide` are used. Rebuild the PDF with `pdflatex main.tex` twice.